PROCESSES FOR THE PREPARATION OF TENAPANOR AND INTERMEDIATES THEREOF

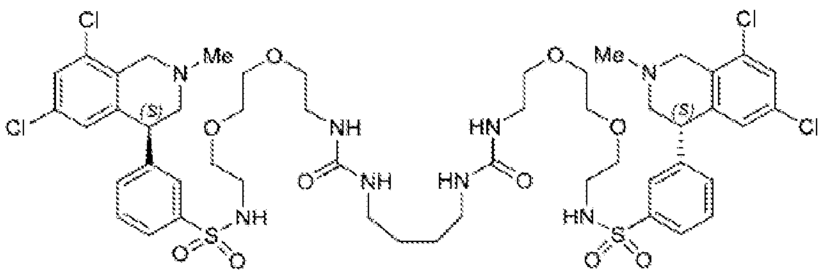
Technical Field

[0001] The present invention provides new procedures and intermediates for the preparation of Tenapanor, Tenapanor dihydrochloride and solid state forms thereof.

Background

[0002] Tenapanor which has the chemical name N,N′-(10,17-dioxo-3,6,21,24-tetraoxa-9,11,16,18-tetraazahexacosane-1,26-diyl)bis(3-(6,8-dichloro-2-methyl-1,2,3,4-tetrahydroisoquinolin-4-yl)benzenesulfonamide) is a selective inhibitor of NHE3 (sodium hydrogen exchanger 3) for the oral treatment of constipation- predominant irritable bowel syndrome (IBS-C) and for the management of fluid overload, dietary sodium restriction and hyperphosphatemia in kidney disease.

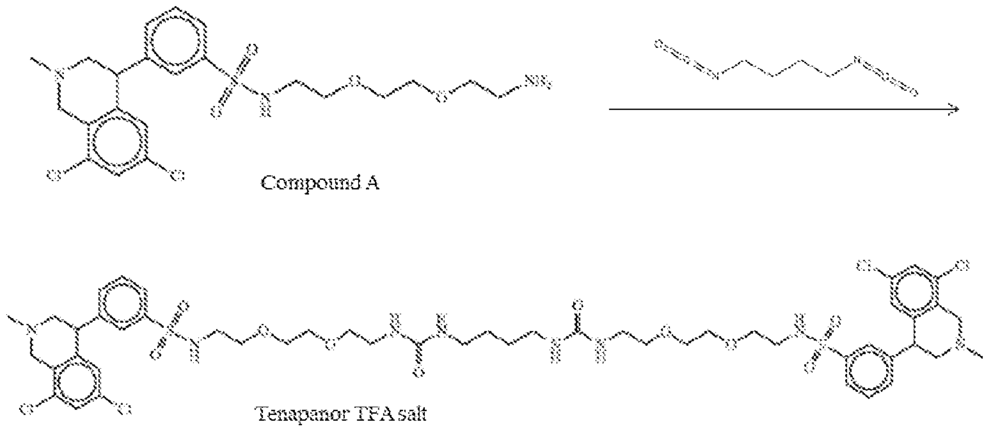
[0003] As described in U.S. Patent No. 8,969,377 and U.S. Patent No. 8,541,448 Tenapanor has the following chemical structure:



[0004] Tenapanor and Tenapanor dihydrochloride are not described to show polymorphs. However, Tenapanor is known to exist in crystalline form and Tenapanor dihydrochloride in an amorphous form, as described in the International Patent Application No. WO 2014/169094. Recently, crystalline Tenapanor base was published in the International Patent Application No. WO 2019/091503.

[0005] Polymorphism, the occurrence of different crystalline forms, is a property of some molecules and molecular complexes. A single molecule, like Tenapanor, may give rise to a variety of polymorphs having distinct crystal structures and physical properties like melting point, thermal behaviors (e.g., measured by thermogravimetric analysis – “TGA”, or differential scanning calorimetry – “DSC”), X-ray diffraction (XRD) pattern, infrared absorption fingerprint, and solid state (13C) NMR spectrum. One or more of these techniques may be used to distinguish different polymorphic forms of a compound.

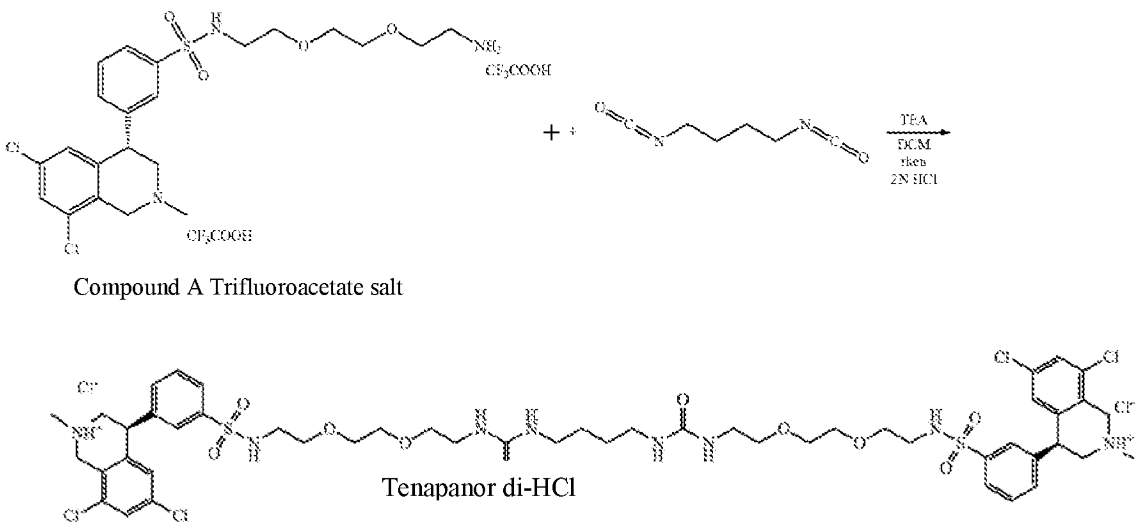
Scheme A:



[0009] The synthetic procedure for preparation of N-(2-(2-(2-aminoethoxy) ethoxy)ethyl)-3-(6,8-dichloro-2-methyl-1,2,3,4-tetrahydroisoquinolin-4-yl) benzenesulfonamide (compound A) is described in U.S. Patent No. 8,969,377 (example 168.2).

[0010] The preparation of Tenapanor di-hydrochloride salt, (S or R)—N,N′-(10,17-dioxo-3,6,21,24-tetraoxa-9,11,16,18-tetraazahexacosane-1,26-diyl)bis(3-((S)-6,8-dichloro-2-methyl-1,2,3,4-tetrahydroisoquinolin-4-yl)benzenesulfonamide)bis-hydrochloride salt, is described in U.S. Patent No. 8,969,377 (example 202) , as demonstrated by the following scheme B.

Scheme B:



[0011] According to scheme B, (S) or (R)-N-(2-(2-(2-aminoethoxy)ethoxy)ethyl)-3-(6,8-dichloro-2-methyl-1,2,3,4-tetrahydroisoquinolin-4-yl)benzenesulfonamide bis(2,2,2-trifluoroacetate), referred to herein as (R) or (S)-Compound A trifluoroacetate salt,